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Digital inclusion challenges in Bangladesh: the case of the National ICT Policy

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ABSTRACT

This study assesses the impact and effectiveness of the National Information and Communications Technology Policy (NIP) in Bangladesh, which is a key legal framework for 'Vision 2021' and 'Digital Bangladesh'. It investigates the extent to which digital inclusion can be achieved by the NIP strategies, as well as exploring the conceptualisation of digital inclusion, which undergirds the contemporary debate on the digital divide and inclusion. The study employs a qualitative approach using policy goal-means analysis to explore critical scrutiny of the internal rationality and logic of the NIP based on three distinct criteria of the digital inclusion analytical framework: ICT access, use and skills. The analysis argues that the policy is ambiguous and techno-centric, with a narrow digitisation frame of reference, which does not comprehensively address the issues associated with digital inclusion. Policy needs to be consistent and relevant to a person's daily life, giving the user a sense of empowerment and skill. The article concludes that a focus on a skill-based approach and the context of societal challenges of Bangladesh is crucial to ensuring digital inclusion.

KEYWORDS

ICT policy; ICT4D; digital inclusion; ICT access; engagement; skills; Bangladesh

Introduction

In the discourse of international development, Information and Communications Technology (ICT) is often claimed as a catalyst for socio-economic development to resolve social problems in developing countries. At the same time, moving toward a 'knowledge-based' and 'digital society' has had economic and social impacts that exacerbate existing inequalities (James 2005; Selwyn 2004). The World Bank (2016, 171) reports that 'digital technologies, too, often fail to empower citizens'. Such a divide can be located across societal levels in many contexts. It creates barriers to employment and escalates social exclusion among low income, rural people and older or disadvantaged groups (Hilbert 2011). This has led to scholarly debates and discussions about the reasons for failure (e.g., Dada 2006; Heeks 2002), including the role of government policies. Moreover, the United Nations Conference on Trade and Development (UNCTAD) reports many developing countries have not yet defined policy strategies (UNCTAD 2014), while some have developed their national plans to accelerate broadband adoption and network connectivity to develop their ICT policies (Duncan-Howell and Lee 2008).

Many South Asian countries like India and Pakistan have implemented ICT policies and digital agendas in order to tackle social problems and encourage economic growth. Both countries have experienced several successes and failures in ICT projects. Palvia, Baqir, and Nemati (2015) find many drawbacks and gaps between policy design and outcome in the ICT policies of Pakistan. In India, analysing the ICT policy of Andhra Pradesh (1994–2002), Dabla (2004) has found that

different ethnic minorities and other social groups remain excluded due to lack of contextual appropriation. Policymakers and social scientists have worried about equal distribution of ICT and Internet access since the inception of the Internet as a mass medium in the mid-1990s (Epstein, Nisbet, and Gillespie 2011). Since 2014, UNCTAD has carried out the ICT Policy review framework (2014) and programmes to identify policies, plans and implementation mechanisms in order to leverage ICT for developing countries (UNCTAD 2018).

As in many other developing countries, the government of Bangladesh introduced several ICT policies, acts and strategies (see, Appendix) to establish a transparent, inclusive and accountable government for economic growth and social development. However, Bangladesh, in the past, has experienced both success and failure of implementation of policy goals (Hasan 2012). Recently, there have been quite a number of government initiatives (such as access to information [a2i]) for implementing ICT driven activities, and the government is planning to amend the existing policy to make it more relevant due to the 'emergence of 5G technology and fourth industrial revolution' (New Age 2018).

This research was driven by the literature on 'ICT policy', 'digital inclusion', 'Information and Communication Technologies for Development' (ICT4D), which has begun questioning and rethinking digital inequalities and development (Avgerou 2010; Dijk 2005; Epstein, Nisbet, and Gillespie 2011; Heeks 2010; Helsper 2008; Warschauer 2004). However, many studies (e.g., Choi et al. 2016; Mansell 2014) argue that little academic attention has addressed the digital inclusion from a policy perspective and the consequence of ICT policies in developing countries. Whereas, existing studies have been conducted on either ICT policy or the government's ICT driven projects, no significant studies have addressed the issue of digital inclusion from the ICT policy perspective. Therefore, it is crucial to evaluate and review the policy critically in order to facilitate future policymaking mechanism and application (Kim, Shin, and Lee 2015).

This article aims to critically evaluate National ICT Policy (NIP) in order to develop better policies that incorporate a broader understanding of digital inclusion beyond ICT access. The theoretical section of this study discusses the relevant literature to conceptualise the idea of digital inclusion; it subsequently explores socioeconomic challenges and the ICT Policy framework to explore the digital context of Bangladesh. Following the Dutch political scientists Gijsbert Kuypers (1980), Van de Graaf and Hoppe (1992), this study employs the method of 'goal-means tree analysis', an approach to qualitative document analysis that allows a detailed methodological process to uncover (mis)-matches between goals and means in media policy and strategy texts (as cited in Vanhaeght 2019). Applied in an ICT policy context, the method is used as a technique to assess ICT policy mechanisms, contextualising the research problem and identifying potential gaps with a visual overview of goals and means of the policy plan. The findings of this study highlight the significance of a more contextualised approach in order to take the social context of digital exclusion into consideration and argues for policy that clarifies goals consistently in order to ensure better outcomes.

Bangladesh: socio-economic structure and challenges

Bangladesh is a relatively small country by geographic area, encompassing 147,570 square kilometres with 162 million residents, making it the world's most densely populated country (BBS 2017). The Asian Development Bank (ADB) reports that, the country's economic progress has been strong over the past decade with a gradual increase in Gross Domestic Product (GDP) expected to reach 7.2% in 2019. The population living below the national poverty line has dropped from 31.5% in 2010 to 24.3% in 2016 (ADB 2018). In addition, Bangladesh has made substantial progress in improving outcomes on the Human Development Indexes and has met several targets of the UN Millennium Development Goals. According to the World Bank report,

Rapid growth enabled Bangladesh to reach the lower middle-income country status in 2015. In 2018, Bangladesh fulfilled all three eligibility criteria for graduation from the UN's Least Developed Countries (LDC) list for the first time and is on track for graduation in 2024. (2019)

However, like many other developing countries, Bangladesh has a high level of inequality along with many social problems. The country continues to have structural constraints on employment, including high levels of income inequality, gender discrimination, overpopulation and political instability (Williams 2018). There are many marginalised and socially excluded groups, such as Hijra continue to experience discrimination due to their gender identity (Aziz and Azhar 2019). The majority of Bangladeshis (66.2% of the population) live in rural areas (BBS 2017). While rapid industrialisation has shifted the country's economy to manufacturing and industry, the economy remains largely dependent on agriculture (Khan 2015). Moreover, Bangladesh is one of the most climate-vulnerable countries in the world intensifying its socioeconomic problems. The World Bank reports (2018) that the country's low-lying and climatic topographies, combined with its continued population growth make it extremely vulnerable to many natural calamities, like floods, droughts, cyclones and earthquakes.

Bangladesh has all the indicators of an underdeveloped political environment where democracy is yet to be institutionalised (Khan 2015). A recent study by Asadullah and Chakravorty (2019) shows that there are fewer signs of progress in economic governance as well as less control in combating bureaucratic corruption at all levels of government. Recent debates about Bangladesh's war of liberation continue to polarise citizens at social, political, and intellectual levels (Islam and Islam 2018). In addition, recent concerns have been raised about emerging authoritarian tendencies and a developing 'culture of fear' as the ruling government imposes surveillance mechanisms and censorship practices (e.g., the Digital Security Act 2018) against journalists and the media (Adams 2019). In this context, the state of governance in Bangladesh can be categorised as an immature polity with an authoritarian regime that is highly influenced by those who have money and political power.

ICT policy context: Bangladesh as a digital society

The government places extensive emphasis on digital technologies as a significant force in the development process. 'Digital Bangladesh' is a state sponsored digital initiative to promote several ambitious ICT projects aiming for a poverty free and more inclusive society by 2021. It is part of an electoral pledge of the ruling party, the *Awami League*, and has become a major political commitment over the last decade (Islam and Grönlund 2011). The phrase Digital Bangladesh has become a 'political tool' that has been often used by the ruling party in relation to any ICT development project. It has frequently been mocked as a 'joke' by the opposition parties and intensively criticised as government propaganda. In this context, certain ICT policies have emerged with the cooperation of international agencies (such as UNDP and USAID) to accelerate the digitisation process as a means to economic growth and social development. The government has initiated the Public-Private Partnership (PPP) model to expedite ICT infrastructure and public services development, and consequently the shares of ICT investment have been increased from around 2% to 6% of the country's Gross domestic product (GSMA 2018).

The National ICT Policy (NIP)¹ has become a flagship regulatory framework for the digital agenda in Bangladesh. As a policy promise, there is an extensive network of 4500 Union Digital Centers (UDC)² across the country designed to ensure rural empowerment and inclusion (ITU 2018). One of the strategic outputs of the digital agenda is the *Access to Information (a2i)* programme, which aims to improve the quality and ease of access of decentralised administrative services to the citizens of Bangladesh (A2I 2009). The NIP offers a long-term plan for ICT as a catalyst for socioeconomic and human development. The long-term strategy for 'Vision 2021' was defined in the Perspective Plan (PP) (see Figure 1), while specific strategies and the task of implementation were articulated through two phases: the Sixth Five Year Plan (6FYP) (2011–2015) and the Seventh Five Year Plan (7FYP) from 2016 to 2020 (Commission 2012).

Recently, the National Economic Council (NEC) has proposed the Second Perspective Plan to be implemented from 2021 to 2041. It aims to eliminate extreme poverty and reach upper middle-income by 2030 and high-income country around 2041 with zero poverty (UNB 2020).

WHAT	Digital Bangladesh	Bring socioeconomic changes through ICT and digital revolution across the country.
HOW	<i>Vision 2021</i>	The goal is to alleviate poverty and become a middle-income country following the strategy of two phases 1 ST Perspective Plan (2011–2015 &) & 2016-2020).
	<i>Vision 2041</i>	The goal is to become upper middle-income country by 2030 and high-income country by 2041. As a strategy, 2nd Perspective Plan (2021-2041) was announced on 25 February 2020.

Figure 1. Government digital development agenda in Bangladesh.

The NIP was first adopted in October 2002, then modified in 2008 and 2009, respectively. Finally, on 5 August 2015, the government revised and amended it with a significant adjustment introducing several strategic themes and action plans. The NIP is designed and formulated by the ICT division under the Ministry of Posts, Telecommunication and Information Technology (MoPTIT).³ The ministry has two divisions that facilitate all major ICT initiatives and digital agenda in Bangladesh, the Posts and Telecommunications Division and the ICT Division. The policy is proposed based on the constitutional framework of Bangladesh that emphasised social inclusion and citizen's equal participation in society.

Bangladesh has reached widespread telecommunications coverage through wireless devices. The GSMA (2018) reports that the 3G (third generation) spectrum currently covers about 93% of the Bangladeshi population's mobile internet usage – the figure having been just 21% in 2017 – while 4G (fourth generation) services were launched in February 2018. However, the country still lags behind compared to other South Asian countries in terms of speed and price. The International Telecommunication Union (ITU) annual report shows Bangladesh has the fifth largest offline population in the world (the country has world's eighth largest population), ranked 147th globally and 30th of Asian countries in the ICT Development Index (ITU 2017). Internet access is eight times lower in rural areas than in urban areas (Alam 2014). Many people remain unable to access ICT due to unstable electricity, remote locations, and lack of awareness of the potential benefits of ICT (Ullah 2016). At the same time, online freelancing has gained much popularity over the last couple of years. According to the Oxford Internet Institute (OII), Bangladesh has been ranked as the second largest online labour supplier country after India (Zaman 2019). Bangladesh has linked around 79,000 km optical fibre with an optic submarine connection network as a part of the 16-country consortium between Southeast Asia, the Middle East and Western Europe 4 (SEA-ME-WE 4)⁴ in 2005 and SEA-ME-WE-5 in 2017 (ITU 2018).

Until the early 1990s, the entire telecommunication sector was under strict government control. A substantial change in the ICT environment was observed after the permission of Nationwide Telecommunication Transmission Network (NTTN) license in 2009 (Hasan 2012). There are several other policies and regulatory frameworks (see, Appendix) that have contributed to ICT market liberation and infrastructure development.

Understanding digital inclusion

As ICTs have evolved, policymakers and academics have addressed many ideas and concepts in order to explore the trends and significance of digital connectivity. Terms like digital inclusion, digital divide and digital participation have been considered in scholarly debates and policymaking process and practices over the past two decades (Dezuanni et al. 2018). However, the idea of digital inclusion is difficult to define as it depends on different socio-economic conditions and types of technology in the context of developed and developing countries (James 2005). While developed countries

focus on high-speed internet access, developing countries only desire physical access. Researchers increasingly argue for a change in definition to go beyond the idea of Internet access, instead focusing on social, psychological and cultural aspects that inhibit digital inclusion (Dijk 2006). Addressing these inequalities and gaps, Dijk categorised four consecutive types of access: motivational, physical, skills and usage. Likewise, other limitations may include ICT literacy, digital skills (Hargittai 2002), and psychological or motivational reluctance (Van Dijk and Hacker 2003).

Selecting suitable ICT tools and indicators to measure digital inclusion requires a comprehensive approach, including several aspects of our everyday lives. Bradbrook and Fisher (2004) propose the '5 Cs' model of digital inclusion; connectivity (access), capability (skill/ability), content, confidence (self-efficacy/motivation), and continuity. Helsper (2008) explains that continuity is related to digital engagement with the Internet and other ICT infrastructure. Likewise, the *Digital Economy and Society Index (DESI)* (EC 2018) reviews digital performance in European Union regions based on six main factors: connectivity, human capital, use of the internet services by citizens, integration of digital technology by businesses, digital public services and research and development of ICT (EC 2018). Thomas et al. (2018) propose three critical aspects of digital inclusion: access, affordability, and digital ability. This study provides an analytical framework with three distinct areas of digital inclusion to evaluate the ICT policy discourse (A) ICT access, (B) ICT use and engagement (the nature and extent of use), and (C) ICT skills.

- (A) *ICT Access*. Recent policy and theoretical debates on digital inclusion have shifted away from physical ICT access. The idea of digital inclusion focuses mainly on lack of access (Dijk 2005). However, it remains ambiguous as to which aspects of access, such as quality, infrastructure or speed, play a significant role in ICT engagement. Drawing on Helsper's conceptual framework (2008), this study defines ICT access as related to digital resources, platforms and connectivity.
- (B) *ICT use and engagement*. There are several ways that an individual may use digital technologies. For example, the Internet and mobile phones are widely seen as crucial indicators of social development and participation. While there is no specific definition for digital engagement, the nature and scope of engagement is directly involved with a choice of activities and technologies. However, many studies define digital engagement as Internet-centred (Helsper 2008). This study defines ICT use and engagement in the context of delivering different types of facilities and extending the scope of digital technologies into different aspects of everyday life.
- (C) *ICT Skills*. This term has been used and has evolved interchangeably with other concepts such as digital competence (Ala-Mutka 2011), ICT literacy (Erstad 2006), digital literacy (Erstad and Gillen 2019), new literacies (Lankshear and Knobel 2011), twenty-first century skills (Binkley et al. 2012), and digital ability (Thomas et al. 2018). All these concepts connote that digital technology requires certain competencies (such as skill, literacy, and motivation). Helsper (2008) divides ICT skills into four broad categories: technical, social, critical and creative skills. Thus, digital exclusion may take place due to lack of training and hands-on experiences.

From digital divide to digital inclusion and ICT4D

Early empirical studies on the digital divide focus on a gap in terms of ownership and physical access to digital devices, which was criticised as being a reductionist approach (Vehovar et al. 2006). The continuous technology gap between developed and developing countries coupled with other inequalities, such as socioeconomic status, education and gender have broadly been defined as the digital divide (James 2005; Selwyn 2004). By the end of the 1990s, a new field known as ICTD or ICT4D (ICT for Development) emerged. ICT4D is premised on the idea that 'ICT can improve the socio-economic conditions in developing countries' (Avgerou 2010, 1). In a recent study, Hussain and Brown (2018) developed a comparative framework for ICT4D initiatives in Bangladesh and Rwanda by adopting a critical realist philosophical approach. Exploring the idea of ICT4D, Avgerou

(2010) classified different theoretical discourses and conceptual arrangements for Information Systems (IS) Innovation into four discourses. These discourses have been framed along with two axes of comparison (see Figure 2). The vertical axis relates to the nature of the ICT transformation while the horizontal axis refers to diffusion and innovation of knowledge, either through an ethno-centric approach or through a socially and culturally embedded knowledge perspective.

Similarly, Heeks (2010) argues that unless a long-term development strategy and implementation plan is carried out, ICT4D would not bring considerable and sustainable impact. In this context, attention in ICT4D research has begun to shift to a more holistic view of development, beyond just economic development (Zheng et al. 2018). Many studies (for example, Avgerou 2010; Bradbrook and Fisher 2004; Dijk 2006) argue that ICT is not a magical solution to complex social problems and may potentially even worsen social inequality (Heeks 2010). The digital divide escalates social exclusion among low-income families, rural people, and older or disadvantaged groups (Hilbert 2011). Recently, focusing on the digital transition in the context of contemporary new media and communication inequalities, a study by *Euromedia Research Group* (Trappel 2019) explores how digital media create a knowledge gap that transforms into digital divides. It depicts how algorithmic modelling, digital surveillance, big data and the internet of things (IoT) are shaping new inequalities as well as the power extraction of expensive industrial corporations in the age of digital media.

The attention within digital inclusion debates has moved from digital divides to different stages of inclusion. Research shows that digital inclusion has been related to social inclusion since the early 2000s (Dezuanni et al. 2018). Dezuanni et al. (2018) address a shift from universal access as the central focus of digital inclusion to the importance of digital literacy as well as awareness of the economic, social and health benefits of ICT. Similarly, Yu et al. (2018) describe digital inclusion as an ideal condition where regulatory agencies ensure ICT for all citizens in order to ensure equal digital participation and the utilisation of available digital resources. Accordingly, Thomas et al. (2018) argue 'Digital inclusion is based on the premise that everyone should be able to make full use of digital technologies ... Social and economic participation lies at the heart of digital inclusion' (5–7). Concerning equitable distribution of ICT initiatives, Notley and Foth (2008) claim that policymakers need to consider social inclusion and social capital.

Finally, to define digital inclusion specifically, it is crucial to recognise the subtle shift from physical access to skills, engagement and participation. In the most developed countries, the divide in physical access appears to be ending, whereas underdeveloped countries are yet to overcome this problem. In this study, digital inclusion is viewed as an interrelated concept that indicates actual conditions of

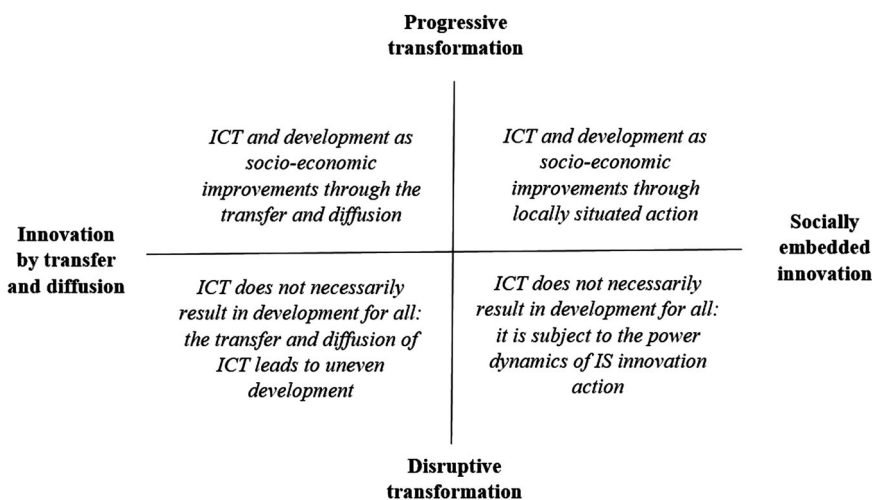


Figure 2. Four discourses on ICT4D. Source: Avgerou (2010).

digital exclusion and divide. Whenever the issue of the digital divide is solved, digital inclusion is consequently ensured. The digital divide continues and perhaps deepens in different environments; therefore, a more systematic and theoretical analysis is required.

Methodology

A policy is a set of instructions – including goals, objectives and other related strategic actions – which often feature as a means-ends relationship. Different objectives or sub goals can be found in a single goal that can be considered as a means or an instrument. The method of goal-means tree analysis is an approach to qualitative document analysis that assesses the relationships among addressed goals, aspirations and actual means in policy text. It is a visualised technique to analyse texts that are often confusedly arranged in an unstructured manner (Vanhaeght and Donders 2017). The method first emerged in the Dutch Commission for Policy analysis in 1970. The Dutch political scientist Gijsbert Kuypers addressed the foundations for this method as a tool for policy development (Kuypers 1980, 67). However, considering a more contextualised approach, Van de Graaf and Hoppe (1992; see in Vanhaeght 2019) criticise this method for ignoring the broader policy context, and subsequently, they revised the method and renamed it as ‘goal-means interpretation’.

The central idea of this method is to identify the final relationships between goals and objectives and means; which are always directed toward desired policy outcomes (Kuypers 1980; Van de Graaf and Hoppe 1992). This method gives a detailed methodological approach to assess whether the policymakers have a comprehensive policy plan to reach their intended policy goals. It gives an explicit critique of the significant policy features while exposing ignored aspects within the policy framework. It also elucidates the internal rationality and logic within policy documents. Nevertheless, some of the main limitations of this method are that it pays more attention to policy content, ignoring the normative ideas, legitimacy and nature of policy problems (Vanhaeght 2019). In addition, all intended goal-means relations in the policy text can never be entirely analysed (Vanhaeght and Donders 2017).

Following the Van de Graaf and Hoppe’s (428) contextualised approach of goal-means interpretation, Vanhaeght (2019) describes the seven steps for construction of a goal-means tree. Accordingly, this study follows a systematic procedure; first, to read the entire policy text to understand the context and identify the relevant texts based on the three criteria (ICT access, use and skills) of the digital inclusion analytical framework. Then, to determine which provisions of the text can be recognised as a means, objective or goal. Afterwards, some of these fragmented goal-means are incorporated with directional arrows using some terms (such as ‘in order to’, ‘with the purpose of’, ‘with tools such as’, ‘through’, ‘by’, ‘to deliver’, ‘to ensure’) and citations from the policy texts along with the page numbers are provided within brackets. The arrows point from means to goal, specifying the goal-means oriented relationships towards the desired outcome in policy. It allows readers to visualise how each means is associated with respective goals or objectives. This method is appropriate to analyse the rationality of policy and examine to what extent the reasoning or argumentation behind the policy is rational and practical.

Results

The National ICT Policy (NIP) strategy is structured in a hierarchical pyramid. It includes a single vision at the top as the ultimate goal while other interlinked parameters (10 objectives, fifty-four strategic themes, and 235 action items) are placed in subsequent layers. The vision and objectives are aligned with the general national goals, and each objective has several strategic themes⁵ (STs) and actions items⁶ (AIs) (MoPTIT 2015). There are ten related objectives to achieve the policy vision, such as social equity, integrity, universal access, education and research, employment generation, strengthening exports, supports to ICTs, healthcare, environment, climate & disaster management, and

productivity (MoPTIT 2015, 7–8). These objectives are covered by a number of STs that present several AIs to be accomplished in the short-term (2016), mid-term (2018) and long-term (2021). Regarding data analysis, it is important to clarify that the policy objectives are associated with a number of goals. In other words, each objective has a number of STs that can be seen as sub-goals while each ST has several AIs, which are considered as a means to achieve the intended objectives and STs of ICT policy.

ICT access

The NIP defines access as to ‘ensure quality telecommunication technology and internet connectivity to all as a public service obligation’ (MoPTIT 2015, 7). ICT access is proposed in one of the policy objectives as ‘universal access’. This objective is outlined along with multiple STs and AIs. The STs for universal access reflect the government’s goal of reaching all citizens of the country through extending ICT infrastructure. It offers Internet Protocol (IP-based) telecommunication systems to ensure wide access at a reasonable cost. To ensure ICT access, some solutions are addressed, including submarine and broadband connectivity. In addition, another objective, social equity, seeks greater participation from users, and access for all including people with special needs (MoPTIT 2015). There are some STs that are placed to achieve this objective that explicitly focus on the inclusion of marginalised people through ICT initiatives. Primarily, it aims ‘to bridge the digital divide and minimize economic disparity for lower income groups, ethnic minorities and persons with disabilities and special needs’ (MoPTIT 2015, 8) (see Figure 3).

To ensure ICT access, several means are presented across the policy text, such as providing access to critical services and information through affordable and reliable rural connectivity via mobile phone. Regarding digital resources and platform to ICT access, the initiation of citizens’ helpdesks in government institutions, development of software, voice content for disabilities and setting up community e-centres are also proposed. Moreover, several conceptual claims are mentioned, such as bridging the digital divide, citizens’ participation, access to critical services, access to critical information, critical government offices, social safety net programme and rural connectivity. However, these claims are presented in a scattered way without clarifying how they are related to ICT access.

ICT use and engagement

Although there is no specific objective that addresses ICT use and engagement, different types of ICT practices and resources are identified in different policy themes and actions under policy objectives. ICT practices mainly aim to increase economic growth, health services, and to tackle the environmental crisis. The policy objectives have significantly focused on ICT engagement within agricultural and environmental issues. For example, an objective outlines ICT usage to facilitate economic growth, including agriculture and small, medium, and micro enterprises (SME) (MoPTIT 2015, 8). Several means are suggested including community network development, the dissemination of agricultural knowledge, as well as innovations through community radio,

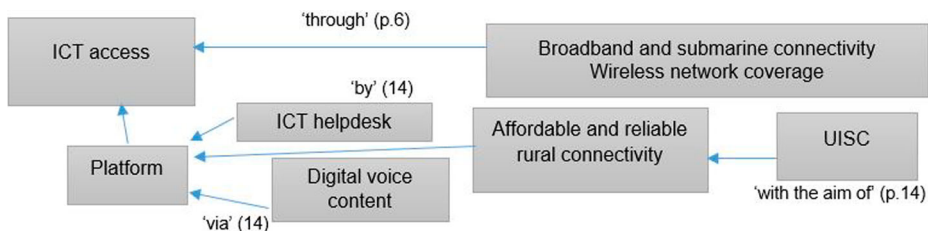


Figure 3. Fragmented goal-means tree of ICT access.

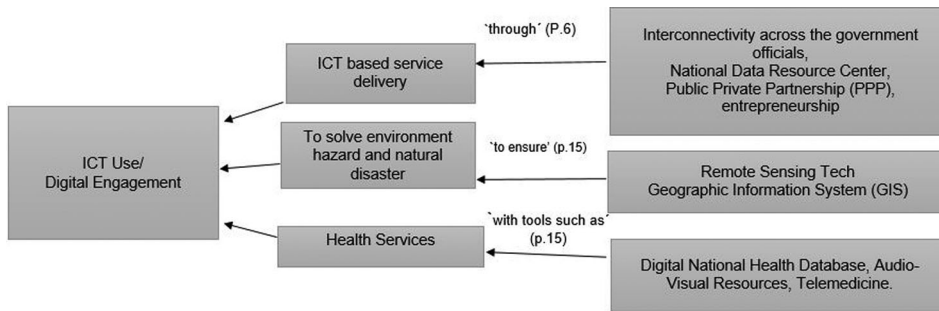


Figure 4. Fragmented goal-means tree of ICT use and engagement.

telecentre and web-based networks. Similarly, health care is one of the crucial applications that has been presented in the policy to ensure quality health care to all citizens (MoPTIT 2015, 8). It offers the development of a national health database system, telemedicine and audio-visual resources as a means to achieve the goal.

Another policy objective ('Integrity') promises ICT use to ensure transparency, efficiency and service delivery to all citizens. In addition to setting up a national data resources centre, the promotion of public-private partnerships (PPP), and of web based and technical mechanisms (such as video conferencing, IP telephony, LAN), are also proposed (see Figure 4).

ICT skills

The policy has no specific objective to describe ICT skills. One policy objective may relate to digital skills and literacy. The objective states that in order to 'ensure computer literacy at all levels of education ... adoption of ICTs [will be needed] through appropriate research and development' (MoPTIT 2015, 7). It offers some approaches, including ICT skills assessment, adoption of ICT education and professional skills, establishing ICT centres, ICT literacy, and courses in primary and secondary education. Moreover, the government has proposed to assess ICT skills during public service entrance exams, creating an imperative for employees in public jobs to learn basic ICT literacy. Similarly, the policy proposes industrial collaborations with universities for ICT research and development of 'market-oriented' ICT skills. However, it does not ensure effective ICT education for all. These multiple strategies are primarily related to ICT infrastructural development, having no specific link to digital skills and literacy improvement (see Figure 5).

Overall, the findings show that the ICT policy mechanism lacks a sound understanding of the interconnected nature of goals-means and their relationship in different levels of the policy process. The major problems related to policy texts are lack of precision, overlapping policy means (such as infrastructural development) and the narrow focus of inclusion (such as access) that made the goals related to digital inclusion inconsistent.

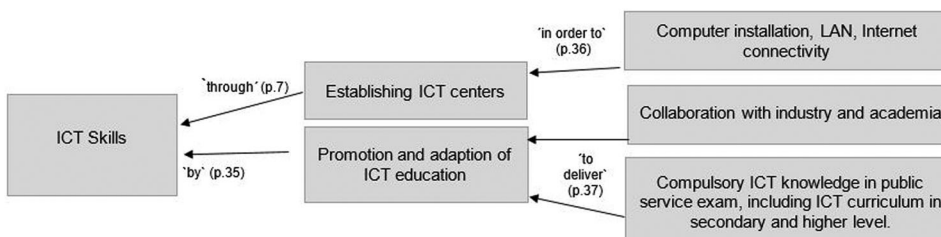


Figure 5. Fragmented goal-means tree of ICT skill.

Discussion and conclusion

This study argues that the NIP is framed without contextualising socio-economic challenges, as well as the local aspects of technical, cultural and human resources. The NIP delivers a general aspiration to ensure access to ICT for all but has no clear reference to poor or socioeconomically disadvantaged groups. First, in relation to ICT access, the policy focuses primarily on infrastructural development across the country. It aims to make rapid expansion of ICT networks and services accessible to all citizens, including socially marginalised groups and communities. The policy suggests several actions to ensure ICT access, such as increasing Internet connectivity and setting up computer laboratories. However, it provides insufficient detail on how this policy will be realised in terms of proposed connectivity and digital resources. Moreover, it frequently focuses on 'affordable accessibility' without stating any action plans towards project implementation and network setup in rural areas, which may lead to a mismatch between policy claims and realities. Similarly, the policy contains vague rhetoric about creating community e-centres for ICT access in rural areas. It is critical to articulate how rural communities will overcome socio economic challenges to utilise these resources.

Secondly, the nature of ICT uses and engagement offered in the NIP may not reach and involve common people, as the policy and strategy formulation remain largely confined within a technocentric approach. ICT scope and nature of digital engagement reflect on government authority and administrative aspects, while the citizens' interests and the scope of people's ICT engagement remain utterly ignored. Moreover, the policy gives a negligible idea of digital scope. Strategies for enhancing the social, cultural, political and civic participation of citizens are explicitly missing in the policy text. Therefore, a more comprehensive and user-centred approach (Helsper 2008) needs to be included within ICT policy strategies.

Thirdly, policy reflects on a market-oriented trend of ICT skills development, which may not have a direct impact on the lives of ordinary and poor people. Inconsistencies are also visible in the ICT policy documents. For example, ICT skills and literacy are addressed using concepts such as 'applied computer' and 'lifelong learning' which need a proper explanation. In Bangladesh, those who are proficient in English language are more likely to get access to a broader spectrum of ICT content than those who solely know Bangla. This linguistic divide, then, is reproduced as a digital divide, especially in remote areas. Although the government has incorporated ICTs into the formal education system, the quality of education varies considerably across the regions, due to the centralised administrative system and poor infrastructure. Moreover, the lack of social security and resources in remote areas creates gender disparity, and thus less access to quality education and study facilities. In this context, the existence of ICT (skills) programmes beyond formal education is crucial in reaching out to diverse groups, and those living in remote parts of the country.

What are the policy challenges?

This analysis argues that ICT policy strategies are formulated without having any specific definition of core concepts to map out the scope and extent of ICT policy. While the definition of basic concepts like ICT is missing, other key issues in the ICT and development discourse, such as 'digital divide', 'e-governance' and 'inclusion' are addressed in multiple policy themes. Furthermore, following each policy objective, STs and AIs are presented in a way that do not give a comprehensive picture for a better understanding of policy goals. Similarly, the policy integrates many topics related to social and economic development in a scattered way. The ambiguity of the policy is reflected in language such as the 'digital divide', which is used without appropriate context and explanation. Several means are mentioned with no proper description, like the idea of 'creating a knowledge-based platform' (which types of knowledge can be shared through which platforms?). At the same time, there are many stakeholders who are involved as primary actors from both the public and the private sector without any proper clarification of the specific tasks that would be carried out at different levels.

In the context of the ICT policy discourse, the government viewed ICT practices as the key catalyst for social change and economic development. This discourse has been incorporated into current ICT policies, which are related to neoliberal and market-driven ideologies (Stevenson 2009). Such a policy trend reflects Avgerou's (2010) first discourse of techno-economic approach on development perception of ICT policy process. Avgerou claims that the idea of setting up digital technologies and internet connectivity appeared as a solution for developing countries. Such computer centred or ICT-based learning in policy settings has little impact on digital inclusion. Byrne and Weilbach (2008) argue that this approach is one of the key barriers to the implementation of ICT projects in developing countries. Besides, the contextual gap which divides goals, policy design and outcomes from policy execution is the common limitation of digital policies in developing countries (Palvia, Baqir, and Nemati 2015). Referring to previous studies (e.g., Haenssger 2017; Hanafizadeh, Khosravi, and Badie 2019), the widespread inactivity and closure of ICT centres shows that new technologies do not equally benefit everyone. In most cases, they do not change the landscape for creating equal opportunity. In the context of Bangladesh, Ullah (2017) claims that ICT projects, such as Union Digital Centres (UDC) have failed to target specifically disadvantaged citizen groups, as the powerless majority of the rural poor did not have access and use of facilities (see also Ullah 2020).

Digital inclusion is actually about social inclusion. The connection between digital and social inclusion has been revealed mainly through government programmes and policies, and it has gradually been shaped through societal contexts and learning environments (Dezuanni et al. 2019). The Australian Digital Inclusion Index (ADII) asserts,

digital inclusion is not just about computers, the internet or even technology. It is about using technology as a channel to improve skills, to enhance the quality of life, to drive education and to promote economic well-being across all elements of society. (ADII 2019)

Furthermore, inclusion can be defined as freedom for all citizens to contribute to social, political, and economic activities in their daily lives (Sen 1999) and allowing people to have control over their destiny (Warschauer 2004). Sen's argument for digital inclusion is to foster 'the ability to access, adapt, and create new knowledge using new information and communication technology' (9). Similarly, Serrano-Santoyo and Rojas-Mendizabal (2017) suggest that non-technology-centric programmes are crucial to achieving sustainable digital inclusion projects, mainly in remote and underserved populations. Thus, the ICT initiatives need to be socially embedded in order to ensure access and utilisation to digital technologies, which lead to Avgerou's *disruptive transformation*.

In conclusion, this article draws the following points from the policy analysis. First, the policy is mainly structured with a narrow digitisation frame of reference (access manners). Such technocentric ICT mechanisms of policy process does not address the comprehensive issues associated with social inclusion, such as inequalities, gender disparity and citizen rights. Next, ICT use and the nature of ICT engagement hardly focus on social involvement. The NIP and policy strategy formulations are limited to access rather than reaching out at the local level. Finally, the propositions on ICT skills and literacy are largely ignored. Therefore, the NIP strategies need a rethink in order to ensure digital inclusion in the context of the societal challenges of Bangladesh. Policy also needs to be consistent and relevant to a person's daily life, giving the user a sense of empowerment and skill.

Acknowledging the methodological limitations, the analysis of ICT policy could be incorporated with other government policies, such as health, education in a comparative manner. In addition, it is required to adopt a mixed method approach in an ICT policy context. For example, an empirical study with 'participatory action research for policy analysis approach' (PAR4P; see Donders and Van Audenhove 2019) combined with experts' interviews, focus groups or stakeholders' consultation would be better comprehended. Similarly, Avgerou's contextualised method would be a useful analytical tool for micro, meso and macro level factors involving different actors and institutions that influence digital inclusion. However, Avgerou claims, 'research that spans micro-macro analytical domains in the social sciences is notoriously difficult' (2010, 12). Further study on ICT policy will forward a theoretical framework with a combination of political economy and the sociology of

ICT-driven economic models. This study hopes to contribute to the government's potential initiatives of policy reform (New Age 2018) in the future, advocating the incorporation of genuine digital inclusion. It offers a significant starting point and a new perspective on both ICT policy mechanisms in Bangladesh and future academic research in this area.

Notes

1. This study considers the latest version of National ICT Policy, 2015 that is published in Bangla language. The policy texts are analysed with additional consultation of NIP 2009, which is written in English.
2. It was previously known as Union Information and Service Centre (UISC).
3. In 2011, the Ministry of ICT was established and later merged with the Ministry of Posts and Telecommunications, and re-named as the MoPTIT in February 2014. The ministry works with the Bangladesh Computer Council (BCC) and Bangladesh Telecommunication Regulatory Commission (BTRC) as key agencies for policy implementation.
4. The system that carries telecommunications between Singapore, Malaysia, Thailand, Bangladesh, India, Sri Lanka, Pakistan, United Arab Emirates, Saudi Arabia, Sudan, Egypt, Italy, Tunisia, Algeria and France.
5. Strategic theme is defined in the policy as 'a broad recommendation to guide the development and implementation of specific action items' (National ICT Policy, MoPTIT 2015, 7).
6. Action item is defined in the policy as 'a particular implementation task under a strategic theme with clearly specified outcomes, time limits and actors' (MoPTIT 2015, 7).

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Appendix 1. The ICT policy initiatives and regulatory framework in Bangladesh.

Policy/Acts	Ministry/Agency	Objectives	Remarks
National Telecommunications Policy (NTP) 1998	MoPTIT and BTRC	To ensure rapid growth of telecommunications technology to ensure socio-economic development.	The policy was first reflected in the Bangladesh Telecommunication Regulating Act (BTRA) of 2001. Later, the policy was revised and developed in 2013.
Bangladesh Telecommunications Act 2001	MoPTIT and BTRC	To form an independent commission to facilitate telecommunication services across the country.	The act was revised and amended in 2006 and 2010 respectively.
ICT Policy 2002	MoPTIT, BTRC, Bangladesh Computer Council (BCC) and Prime Minister's Office (PMO)	To promote and facilitate use of ICT in all sectors and ensure transparent and efficient governance.	The ICT policy was reviewed in 2008 and amended in 2009 with the 'vision 2021' and 'digital Bangladesh'. Finally, in 2015, the policy was revised along with a comprehensive framework of three phases; short term (2016), mid-term (2018) and long term (2021).
Copyright Act 2000	Ministry of Law, Justice and Parliamentary Affairs	To ensure intellectual property rights for creative industries and other related modes of electronic communication.	The act repealed the Copyright Ordinance 1962 and was last amended in 2005. According to this Act, copyright registration is voluntary and not obligatory in Bangladesh. The Act is similar to the Bern Convention and TRIPS Agreement.
ICT Act 2006	Ministry of Law, Justice and Parliamentary Affairs	To ensure legal recognition and security of information and communication technology	The Act was revised and amended in 2009, 2013 and 2016 respectively. The act is notoriously known for article 57. It is most contested and controversial law in Bangladesh regarding freedom of expression and press freedom and political oppression.
Right to Information (RTI) Act 2009	Ministry of Law, Justice and Parliamentary Affairs	To provide legal power to citizens to seek public service-related information	The Information Commission was established under the RTI Act on 1st July 2009. The Commission is a legislative independent body

(Continued)

Continued.

Policy/Acts	Ministry/Agency	Objectives	Remarks
National Broadband Policy 2009	MoPTIT and BTRC	To ensure easy accessibility to the Internet and effective service delivery	formed based on the constitution's fundamental rights mentioned in Article 39. The policy was not successful and remained ineffective. A draft was proposed in 2015 but not finalised yet.
International Long-Distance Telecommunications Services (ILDTS) Policy 2010	MoPTIT and BTRC	To facilitate low-cost service including VOIP services considering the vision of Digital Bangladesh	The policy was drafted in 2007 and remained mostly inactive.
Bangladesh Hi-tech Park Authority Act 2010	MoPTIT, BTRC, other miniseries, like finance, Industries, Law Justice	To leverage hi-tech and software industries in Bangladesh	The act was revised in 2014. The term 'park' relates the biotech, renewable energy and green technology Park. In 2017 Sheikh Hasina Software Technology Park in Jessore and Hi-tech Park in Kaliakoir of Gazipur were built.
Digital Security Act 2016	Ministry of Law, Justice	the legal recognition and security of information and communication technology	Most contested and controversial act similar to ICT Act 2006 and the section 57 of the ICT Act has returned as the section 32 of Digital Security Act. Although the act was revised in 2018, it has been described as draconian by rights activists.

Source: Bangladesh ICT policies and ministries. Compiled by the author.